

## Master Java in 30 Days – Day-by-Day Plan

### WEEK 1: Java Basics & Setup

Goal: Get comfortable with Java syntax, setup, and basic programming concepts.

- **Day 1:** Install JDK, Setup IDE (IntelliJ or Eclipse), First Hello World program
  - **Day 2:** Variables, Data Types, and Type Casting
  - **Day 3:** Operators (Arithmetic, Logical, Comparison)
  - **Day 4:** Input & Output in Java (`Scanner`, `System.out`)
  - **Day 5:** Conditional Statements (`if`, `else`, `else if`, `switch`)
  - **Day 6:** Loops (`for`, `while`, `do-while`)
  - **Day 7:** Practice Day – 5 beginner-level problems (loops + conditions)
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### WEEK 2: Object-Oriented Programming (OOP)

Goal: Master the fundamentals of OOP in Java.

- **Day 8:** Classes & Objects + How Java Works
  - **Day 9:** Constructors & the `this` keyword
  - **Day 10:** Method Overloading & Static Keyword
  - **Day 11:** Inheritance & `super` Keyword
  - **Day 12:** Method Overriding & `final` keyword
  - **Day 13:** Access Modifiers (`public`, `private`, `protected`, default)
  - **Day 14:** Practice Day – Build a basic class hierarchy (e.g., Animal → Dog, Cat)
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Goal: Learn real-world utilities of Java + essential Data Structures

- **Day 15:** Abstract Classes & Interfaces
  - **Day 16:** Exception Handling (`try`, `catch`, `finally`, `throw`, `throws`)
  - **Day 17:** Java Arrays (1D, 2D), Array Operations
  - **Day 18:** String Handling & StringBuilder
  - **Day 19:** Collections - `ArrayList`, `LinkedList`, `Vector`
  - **Day 20:** Collections - `HashMap`, `TreeMap`, `HashSet`
  - **Day 21:** Practice Day – Small project using collections (e.g., Contact Book)
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### **WEEK 4: Projects + Interview Prep**

Goal: Apply knowledge in projects + prepare for interviews

- **Day 22:** Project Planning – Choose your mini project idea
- **Day 23:** Start Project 1 – Bank Management System (CLI)
- **Day 24:** Finish Bank Project (Add features: deposit, withdraw, check balance)
- **Day 25:** Start Project 2 – Student Management System
- **Day 26:** Finish Student Project (Add search, update, delete features)
- **Day 27:** Java 8 Features – Lambdas, Streams (Basics)
- **Day 28:** DSA Concepts in Java – Arrays, Stack, Queue (theory + code)
- **Day 29:** DSA – LinkedList, Tree basics (with Java)
- **Day 30:** Final Revision + 10 Interview Questions + Resume Ready!